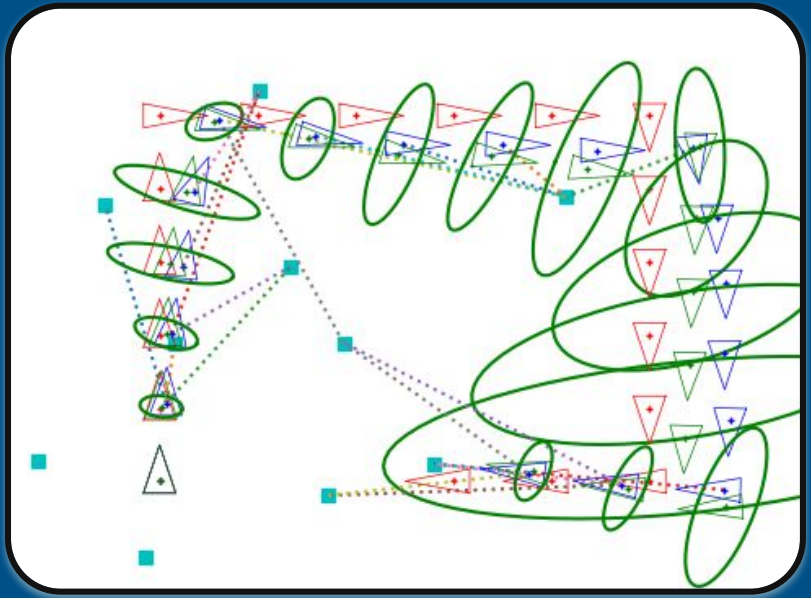
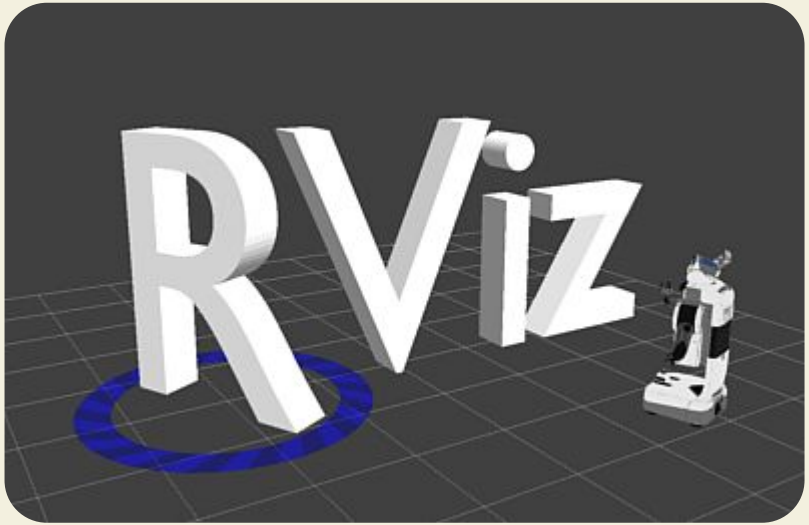
I402: Principios de la Robótica Autónoma

Clases Prácticas
Tutorial 4



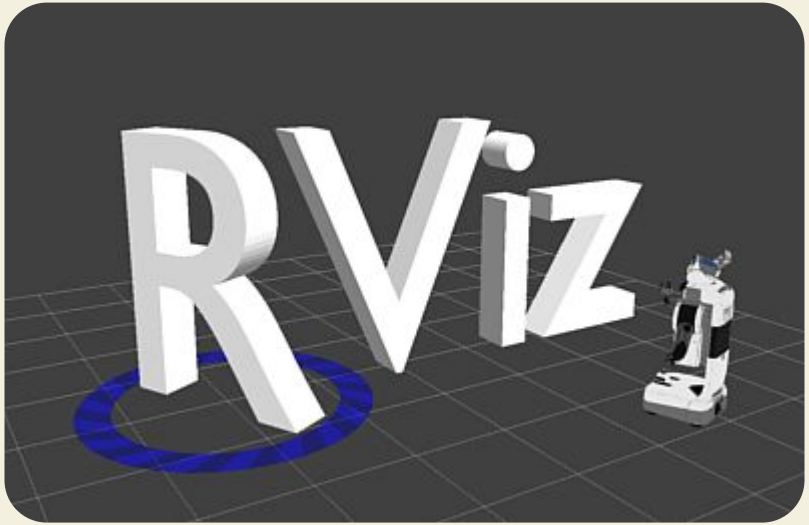
Universidad de SanAndrés

RVIZ



GAZEBO

Simulador



Visualizador

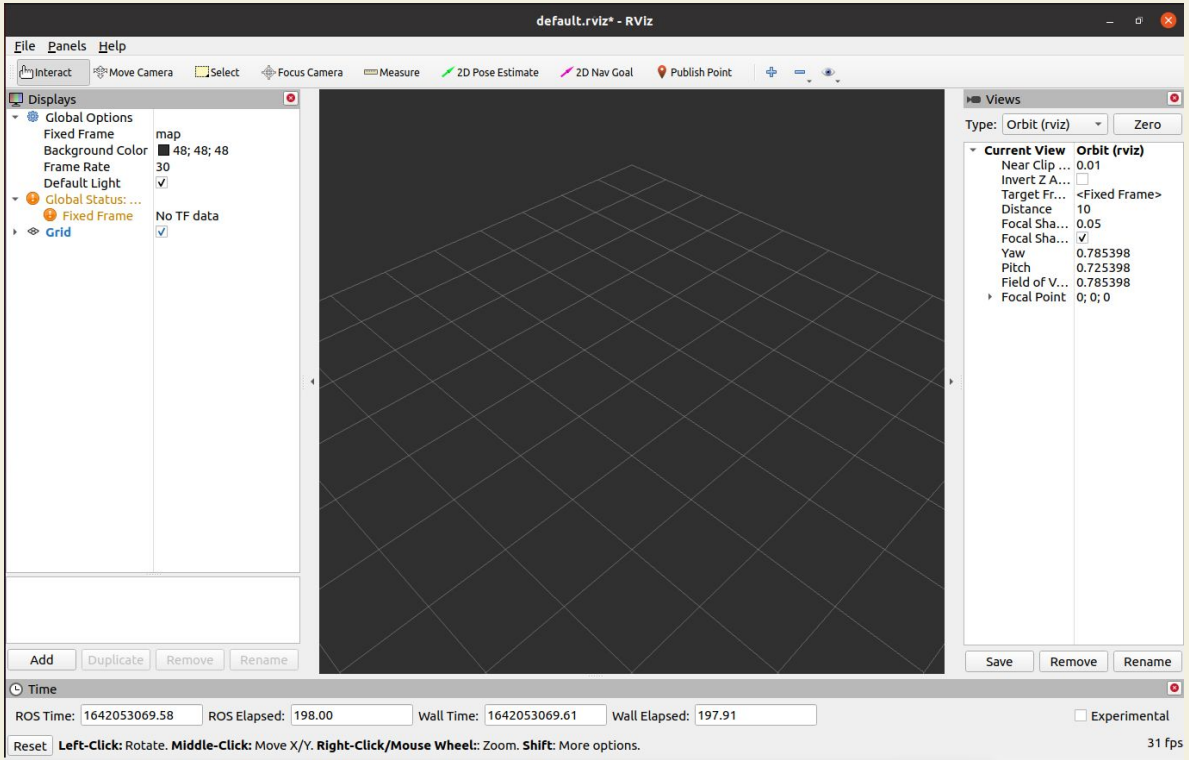
Inicializar

Opción 1

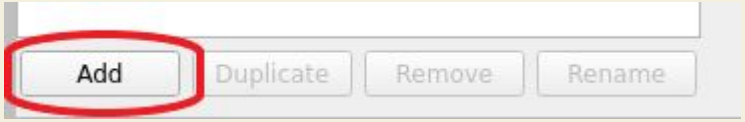
```
rviz2
```

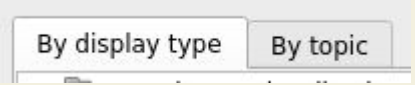
Opción 2:

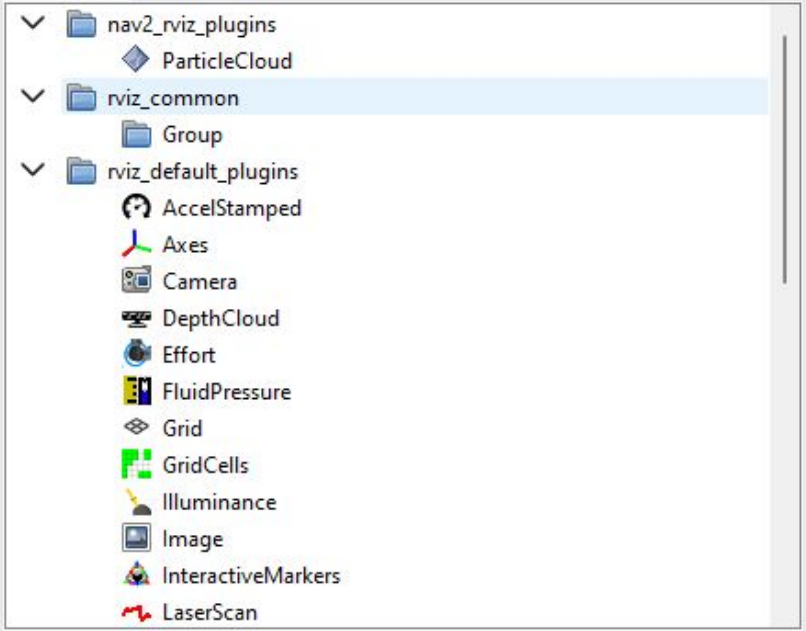
```
ros2 run rviz2 rviz2
```

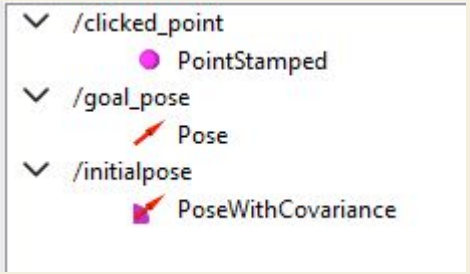


Displays



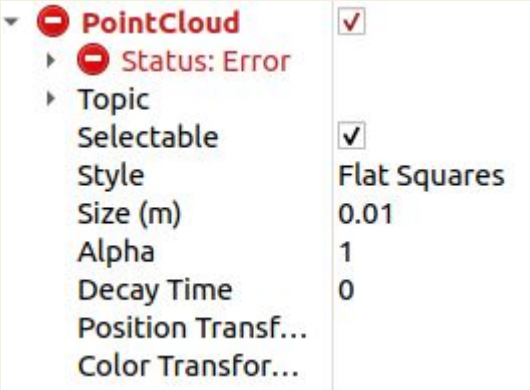






¡Tienen que ser del mismo tipo de msg para que pueda representarse!

Displays y Status



Tipos de views

1. Orbit
2. FPS
3. Top-Down
4. XY orbit
5. Third person

Tipos de views

1. Orbit
2. FPS
3. Top-Down
4. XY orbit
5. Third person

Tipos de Frames

1. Fixed Frame
2. Target Frame

Agreguemos el recorrido del robot

Agreguemos el recorrido del robot

nav_msgs/msg/Odometry Message

File: nav_msgs/msg/Odometry.msg

Raw Message Definition

```
# This represents an estimate of a position and velocity in free space.

# The pose in this message should be specified in the coordinate frame given by
header.frame_id

# The twist in this message should be specified in the coordinate frame given by
the child_frame_id


# Includes the frame id of the pose parent.

std_msgs/Header header

# Frame id the pose points to. The twist is in this coordinate frame.

string child_frame_id

# Estimated pose that is typically relative to a fixed world frame.

geometry_msgs/PoseWithCovariance pose

# Estimated linear and angular velocity relative to child_frame_id.

geometry_msgs/TwistWithCovariance twist
```

Agreguemos el recorrido del robot

nav_msgs/msg/Odometry Message

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header.frame_id

# The twist in this message should be specified in the coordin
the child_frame_id


# Includes the frame id of the pose parent.

std_msgs/Header header

# Frame id the pose points to. The twist is in this coordinate

string child_frame_id

# Estimated pose that is typically relative to a fixed world f

geometry_msgs/PoseWithCovariance pose

# Estimated linear and angular velocity relative to child_fram

geometry_msgs/TwistWithCovariance twist
```

std_msgs/msg/Header Message

File: std_msgs/msg/Header.msg

Raw Message Definition

```
# Standard metadata for higher-level stamped data types.

# This is generally used to communicate timestamped data

# in a particular coordinate frame.


# Two-integer timestamp that is expressed as seconds and nanoseconds.

builtin_interfaces/Time stamp

# Transform frame with which this data is associated.

string frame_id
```

Agreguemos el recorrido del robot

nav_msgs/msg/Odometry Message

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string child_frame_id

# Estimated pose that is typically relative to a fixed world frame.

geometry_msgs/PoseWithCovariance pose

# Estimated linear and angular velocity relative to child_frame_id.

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```

Agreguemos el recorrido del robot

nav_msgs/msg/Odometry Message

File: nav_msgs/msg/Odometry.msg

Raw Message Definition

```
# This represents an estimate of a position

# The pose in this message should be specifi
header.frame_id

# The twist in this message should be specif
the child_frame_id


# Includes the frame id of the pose parent.

std_msgs/Header header

# Frame id the pose points to. The twist is

string child_frame_id

# Estimated pose that is typically relative

geometry_msgs/PoseWithCovariance pose

# Estimated linear and angular velocity relative to child_frame_id.

geometry_msgs/TwistWithCovariance twist
```

geometry_msgs/PoseWithCovariance Message

File: geometry_msgs/PoseWithCovariance.msg

Raw Message Definition

```
# This represents a pose in free space with uncertainty.

Pose pose

# Row-major representation of the 6x6 covariance matrix
# The orientation parameters use a fixed-axis representation.
# In order, the parameters are:
# (x, y, z, rotation about X axis, rotation about Y axis, rotation about Z axis)
float64[36] covariance
```

Agreguemos el recorrido del robot

geometry_msgs/msg/Pose Message

File: geometry_msgs/msg/Pose.msg

Raw Message Definition

```
# A representation of pose in free space, composed of position and orientation.


Point position
Quaternion orientation
```

nCovariance Message

ovariance.msg

```
# This represents a pose in free space with uncertainty.

Pose pose

# Row-major representation of the 6x6 covariance matrix
# The orientation parameters use a fixed-axis representation.
# In order, the parameters are:
# (x, y, z, rotation about X axis, rotation about Y axis, rotation about Z axis)
float64[36] covariance
```

```
# Includes the frame id of the pose parent.

std_msgs/Header header

# Frame id the pose points to. The twist is

string child_frame_id

# Estimated pose that is typically relative

geometry_msgs/PoseWithCovariance pose

# Estimated linear and angular velocity relative to child_frame_id.

geometry_msgs/TwistWithCovariance twist
```

Agreguemos el recorrido del robot

geometry_msgs/msg/Pos

File: geometry_msgs/msg/Pose.m

Raw Message Definition

```
# A representation of pose in free space, compo


Point position
Quaternion orientation
```

```
# Includes the frame id of the pose parent.

std_msgs/Header header

# Frame id the pose points to. The twist is

string child_frame_id

# Estimated pose that is typically relative

geometry_msgs/PoseWithCovariance pose

# Estimated linear and angular velocity relative to chi

geometry_msgs/TwistWithCovariance twist
```

```
# This r

Pose pos

# Row-ma
# The or
# In orc
# (x,
float64
```

geometry_msgs/msg/Point Message

File: geometry_msgs/msg/Point.msg

Raw Message Definition

```
# This contains the position of a point in free space

float64 x
float64 y
float64 z
```

geometry_msgs/msg/Quaternion Message

File: geometry_msgs/msg/Quaternion.msg

Raw Message Definition

```
# This represents an orientation in free space in quaternion form.


float64 x 0
float64 y 0
float64 z 0
float64 w 1
```

Agreguemos el recorrido del robot

nav_msgs/msg/Path Message

File: nav_msgs/msg/Path.msg

Raw Message Definition

```
# An array of poses that represents a Path for a robot to follow.

# Indicates the frame_id of the path.
std_msgs/Header header
# Array of poses to follow.
geometry_msgs/PoseStamped[] poses
```

Agreguemos el recorrido del robot

nav_msgs/msg/Path Message

File: nav_msgs/msg/Path.msg

Raw Message Definition

```
# An array of poses that represents a P

# Indicates the frame_id of the path.
std_msgs/Header header
# Array of poses to follow.
geometry_msgs/PoseStamped[] poses
```

geometry_msgs/msg/PoseStamped Message

File: geometry_msgs/msg/PoseStamped.msg

Raw Message Definition

```
# A Pose with reference coordinate frame and timestamp

std_msgs/Header header
Pose pose
```

Launch File

1402

Launch file

```
cd ~/ros2_ws/src/example_package
mkdir launch
cd launch
touch example_launch.py
```

Launch file

```python
from launch import LaunchDescription
from launch_ros.actions import Node

def generate_launch_description():
    talker_node = Node(
            package='example_package',
            namespace='',
            executable='talker',
            name='talker'
        )

    listener_node = Node(
            package='example_package',
            namespace='',
            executable='listener',
            name='listener'
        )

    return LaunchDescription([
        talker_node,
        listener_node,
    ])
```

https://docs.ros.org/en/humble/Tutorials/Intermediate/Launch/Launch-Main.html

Launch file

```python
from launch import LaunchDescription
from launch_ros.actions import Node

def generate_launch_description():
    talker_node = Node(
            package='example_package',
            namespace='',
            executable='talker',
            name='talker'
        )

    listener_node = Node(
            package='example_package',
            namespace='',
            executable='listener',
            name='listener'
        )

    return LaunchDescription([
        talker_node,
        listener_node,
    ])
```

\<namespace>\chatter

https://docs.ros.org/en/humble/Tutorials/Intermediate/Launch/Launch-Main.html

package.xml

```
<exec_depend>ros2launch</exec_depend>
```

setup.py

```
import os

...

data_files=[
    ...
    (os.path.join('share', package_name),['launch/example_launch.py']),
],
```

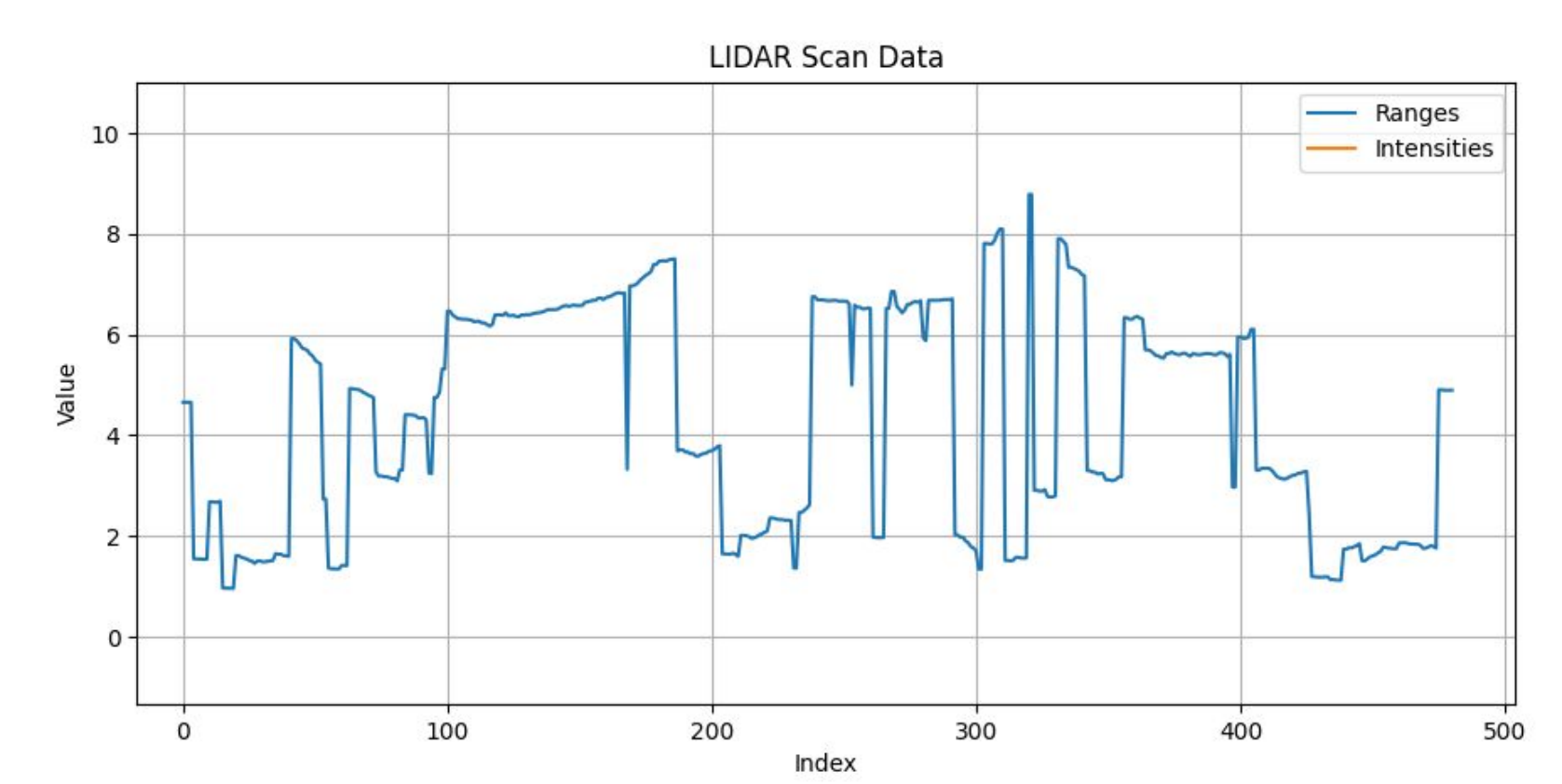
Colcon build

```
colcon build --symlink-install
```

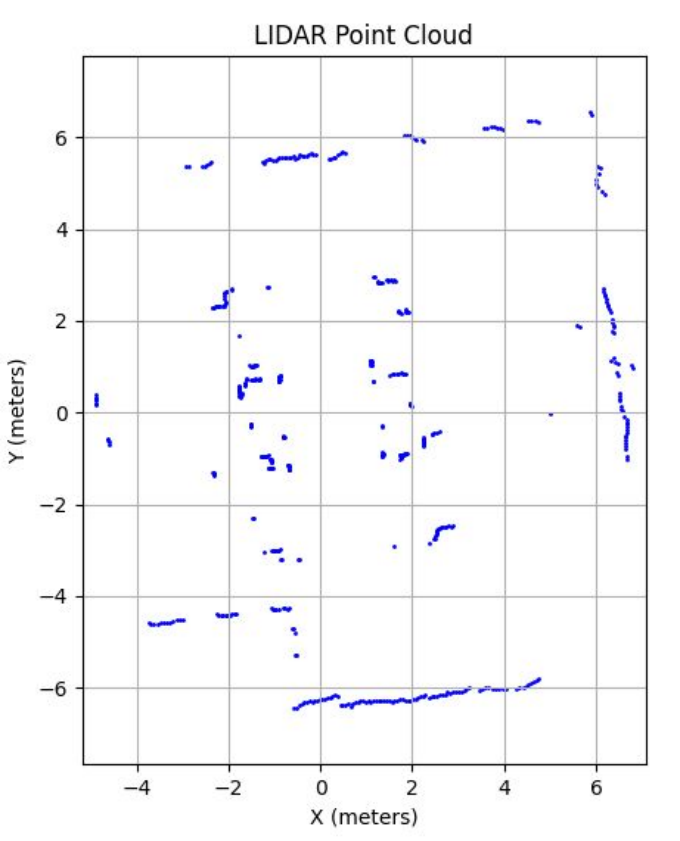
```
source install/setup.bash
ros2 launch example_package example_launch.py
```

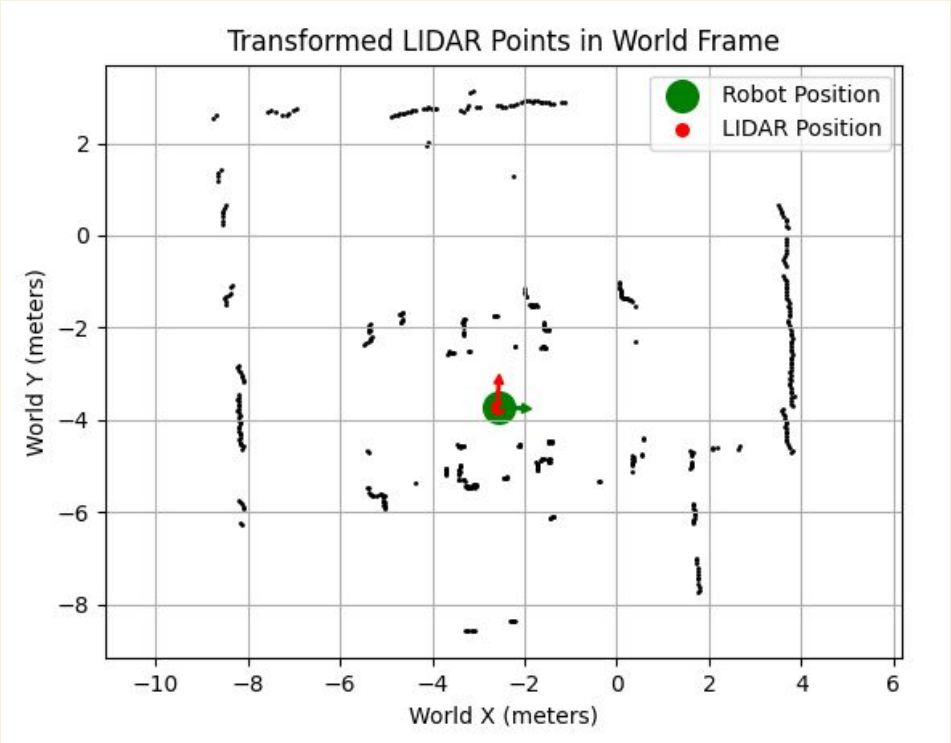
Presentación de TP1 y Discusión de TP2

TP1: Sensado - Resultados

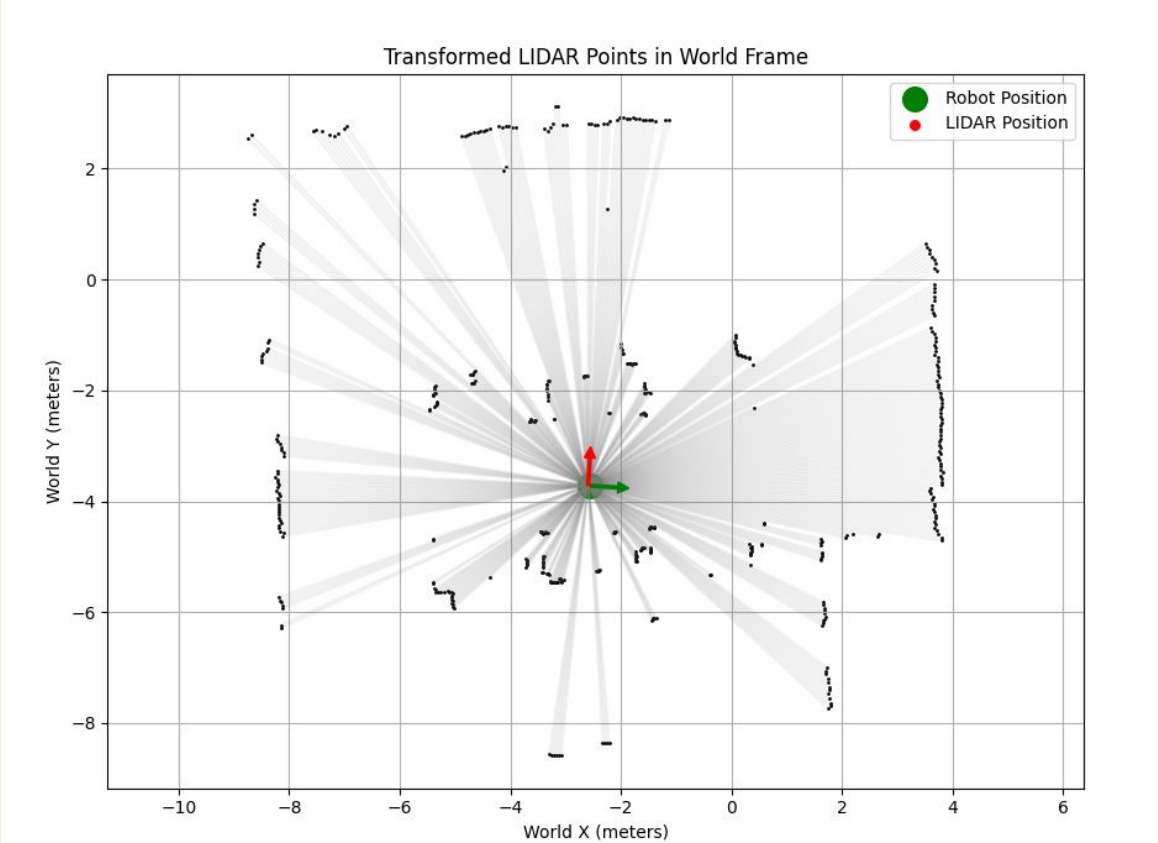


TP1: Sensado - Resultados

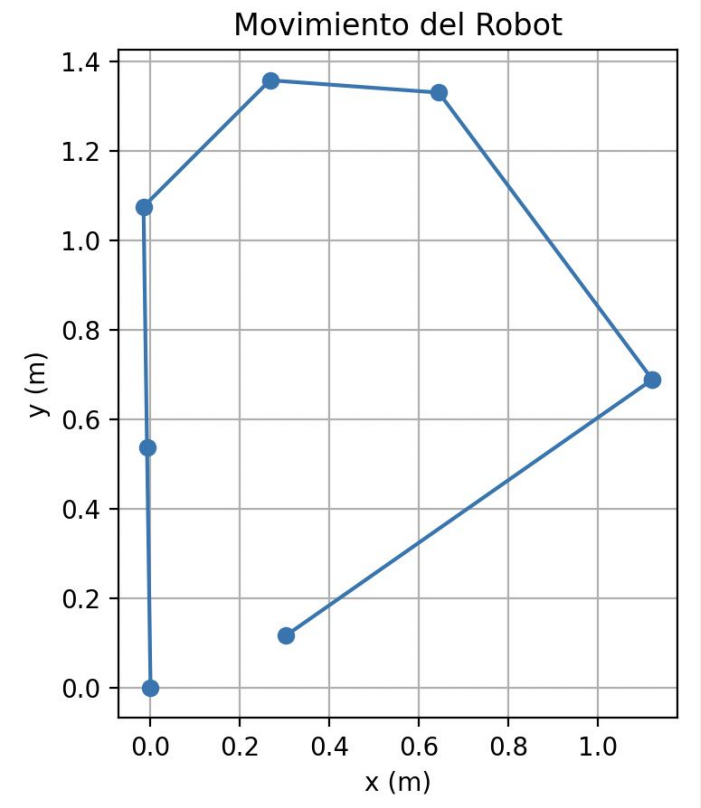




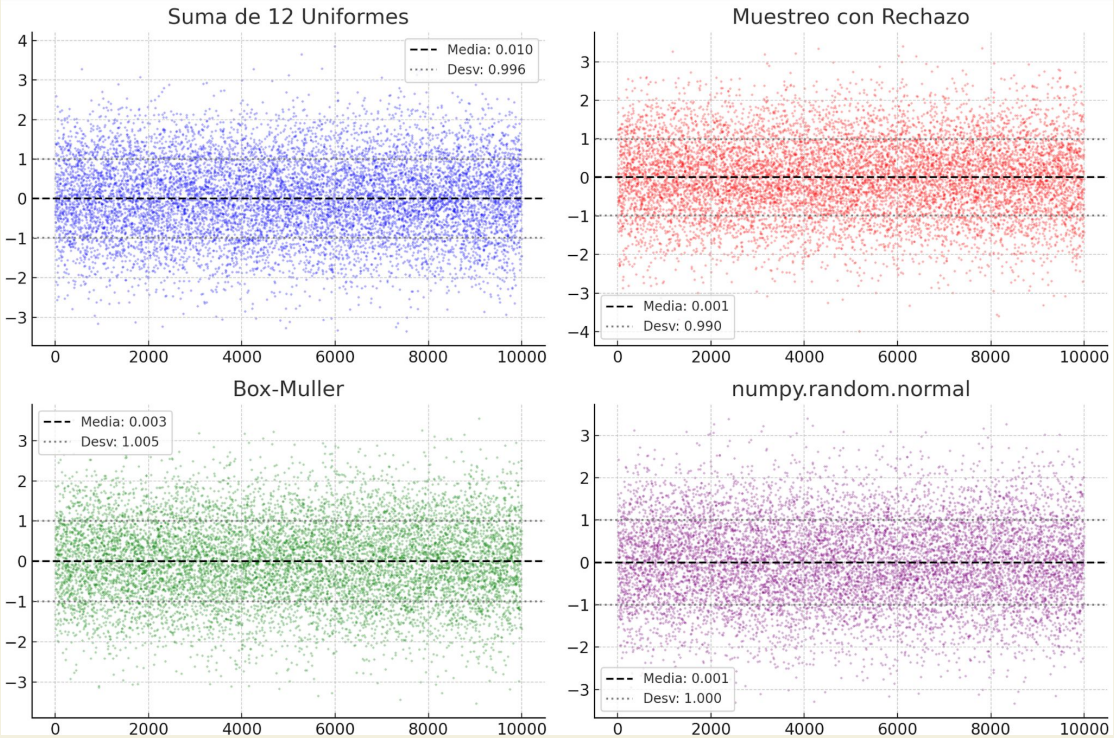
TP1: Sensado - Resultados



TP1: Accionamiento Diferencial – Resultado

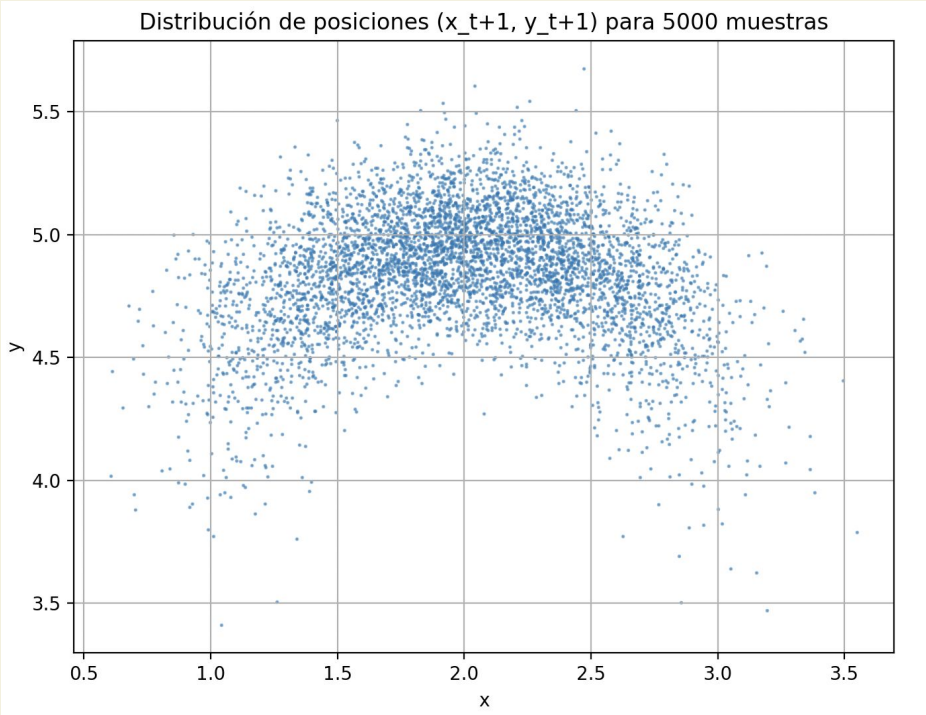


TP2: Muestreo - Expectativas



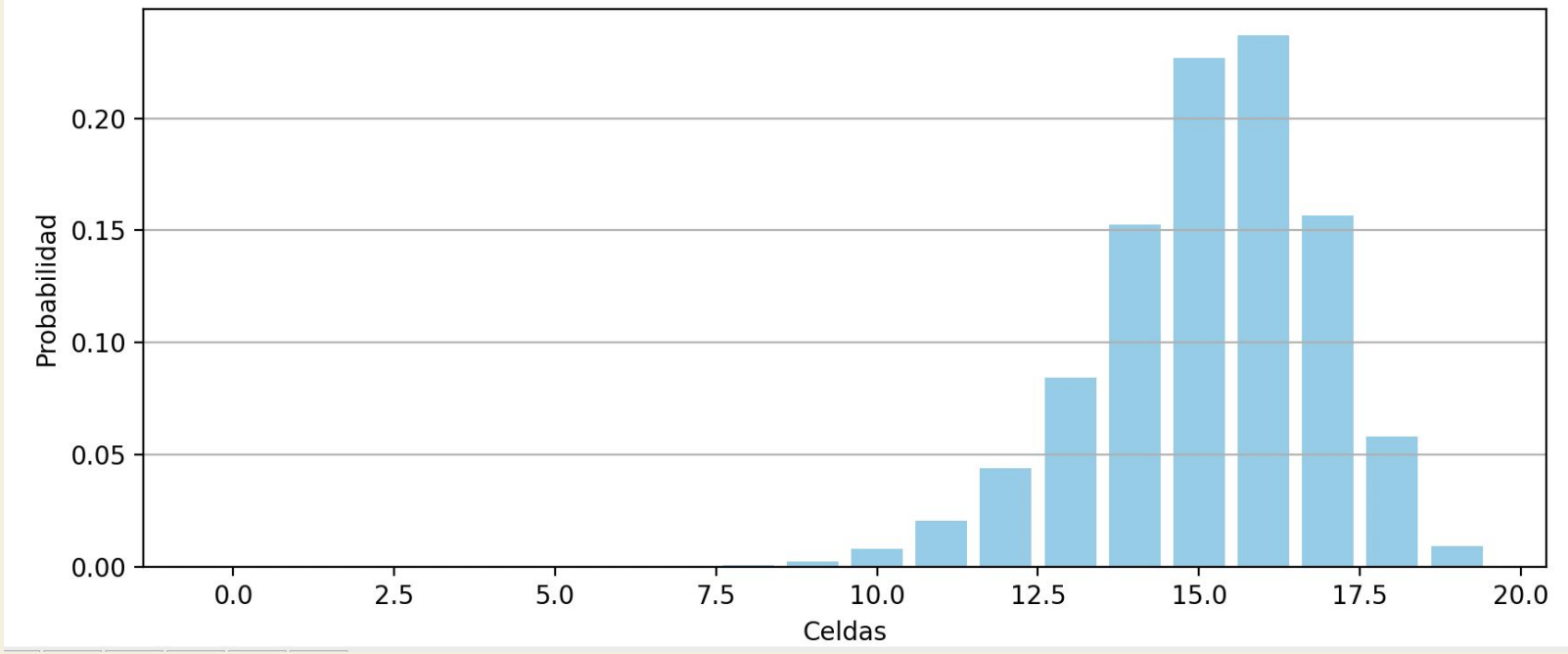
- Cómo se comparan con una distribución normal real? Qué ventajas presentan estos modelos computacionalmente?

TP2: OdomModel - Expectativas



- Qué está ocurriendo físicamente y qué está interpretando mi algoritmo en consecuencia?

TP2: Filtro Discreto - Expectativas



- Qué representan estas probabilidades? A qué instante corresponde?